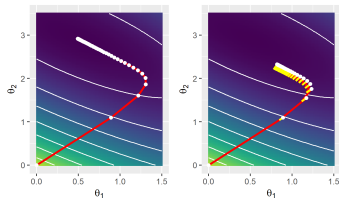
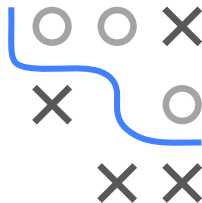


Introduction to Machine Learning

Regularization

Weight Decay and L2



Learning goals

- $L2$ regularization with GD is equivalent to weight decay
- Understand how weight decay changes the optimization trajectory

WEIGHT DECAY VS. L2 REGULARIZATION

Let's optimize $L2$ -regularized risk of a model $f(\mathbf{x} \mid \boldsymbol{\theta})$

$$\min_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \frac{\lambda}{2} \|\boldsymbol{\theta}\|_2^2$$

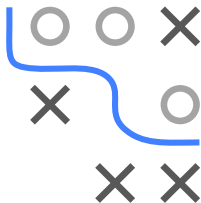
by GD. The gradient is

$$\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \boldsymbol{\theta}$$

We iteratively update $\boldsymbol{\theta}$ by step size α times the negative gradient

$$\begin{aligned} \boldsymbol{\theta}^{[\text{new}]} &= \boldsymbol{\theta}^{[\text{old}]} - \alpha \left(\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]}) + \lambda \boldsymbol{\theta}^{[\text{old}]} \right) \\ &= \boldsymbol{\theta}^{[\text{old}]} (1 - \alpha \lambda) - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]}) \end{aligned}$$

We see how $\boldsymbol{\theta}^{[\text{old}]}$ decays in magnitude – for small α and λ – before we do the gradient step. Performing the decay directly, under this name, is a very well-known technique in DL - and simply $L2$ regularization in disguise (for GD).



CAVEAT AND OTHER OPTIMIZERS

Caveat: Equivalence of weight decay and $L2$ only holds for (S)GD!

- ▶ Hanson and Pratt 1988 originally define WD “decoupled” from gradient-updates $\alpha \nabla_{\theta} \mathcal{R}_{\text{emp}}(\theta^{[\text{old}]})$ as
$$\theta^{[\text{new}]} = \theta^{[\text{old}]}(1 - \lambda') - \alpha \nabla_{\theta} \mathcal{R}_{\text{emp}}(\theta^{[\text{old}]})$$
- This is equivalent to modern WD/ $L2$ (last slide) using reparameterization $\lambda' = \alpha \lambda$
- Consequence: if there is optimal λ' , then optimal $L2$ penalty is tightly coupled to α as $\lambda = \lambda' / \alpha$ (and vice versa)
- ▶ Loshchilov and Hutter 2019 show no equivalence of $L2$ and WD possible for adaptive methods like Adam (Prop. 2)
- In many cases where SGD+ $L2$ works well, Adam+ $L2$ underperforms due to non-equivalence with WD
- They propose a variant of Adam decoupling WD from gradient updates (AdamW), increasing performance over Adam+ $L2$

